

The association between screen time and cognitive function in children: The partially mediating role of putamen volume

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ABSTRACT

Objective: This study investigates the association between screen time and cognitive function and examines whether putamen volume, which decreases with increased screen time, mediates this relationship.

Methods: Using data from the Adolescent Brain Cognitive Development study of nearly 10,000 children aged 9–10 years, screen time was assessed through self-report questionnaires, and cognitive function—including working memory, processing speed, language, executive function, and episodic memory—was measured using the National Institutes of Health toolbox. Linear mixed-effects models were used to analyze the association between screen time and cognitive function, and mediation effects of putamen volume were evaluated.

Results: Screen time was significantly negatively associated with all domains of cognitive function. The volume of putamen partially mediated the relationship between screen time and executive function ($\beta = -0.015$, false discovery rate [FDR] $p = .027$), processing speed ($\beta = -0.012$, FDR $p = .027$), and language ($\beta = -0.005$, FDR $p = .027$).

Conclusion: Excessive screen time was negatively associated with cognitive function in children, with putamen volume involving in the association of screen time with executive function, language and processing speed. These findings could inform strategies to promote healthy screen habits and support cognitive function and brain development in children.

1. Introduction

Screen time among children has increased globally since the COVID-19 pandemic (Madigan et al., 2022; Trott et al., 2022). This phenomenon has prompted researchers to investigate the potential consequences of excessive screen time on cognitive health and neurological development. The rapid proliferation of digital screens in daily life has raised concerns about their potential impact on mental health, behavior, and brain development (Marciano et al., 2021; Paulich et al., 2021; Xie et al., 2020). Spending more time on screens is associated with deteriorating mental health, increased behavioral problems, being overweight, and poor sleep patterns in late childhood (Moradi et al., 2016; Paulich et al., 2021).

A large number of previous studies have examined the relationship between screen time and cognitive function in children (Adelantado-Renau et al., 2019; Mallawaarachchi et al., 2024; Yang et al., 2024).

Adelantado-Renau et al. (2019) conducted a meta-analysis of 30 studies examining the relationship between screen time and academic performance in children and adolescents. It found that television viewing and video game time were inversely associated with academic performance scores, whereas total screen time was not (Adelantado-Renau et al., 2019). In preschoolers, excessive screen time is associated with executive dysfunction, which may mediate the association between problematic behavior and screen time (Likhitweerawong et al., 2024). However, one meta-analysis focusing on children under 6 found there was no statistically significant association in the relation between screen time use and executive functions (Bustamante et al., 2023), while another systematic review and meta-analysis found that the more program viewing and background television were associated with poorer cognitive outcomes (Mallawaarachchi et al., 2024). Some studies have also suggested that moderators, such as the context of screen use and the type of screen devices, may influence the relationship between screen

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time and cognitive function in early childhood (Adelantado-Renau et al., 2019; Yang et al., 2024). For late childhood, spending more time on screens is associated with lower academic performance (Mundy et al., 2020; Paulich et al., 2021) and lower cognitive performance (Walsh et al., 2020) in 9- and 10-year-old children. These findings suggested a negative association between screen time and cognitive function.

Nevertheless, there is a paucity of studies on the mediating effects of brain structure on the relationship between screen time and cognitive function. One previous preschool-based study found a link between increased screen time and brain morphological abnormalities in areas supporting emergent literacy and language (Hutton et al., 2020), suggesting that the brain actively mediates this relationship. Notably, our previous study demonstrated that enhanced screen time was associated with reduced putamen volume in a large sample of children, using data collected from the Adolescent Brain Cognitive Development (ABCD) study (Shou et al., 2025). The putamen, a component of the striatum, is involved in various functions, including language processing, reward regulation, cognition, and addiction (Ghandili & Munakomi, 2025). Previous research has reported that screen exposure is associated with altered functional connectivity between the frontoparietal network and the putamen (Chen et al., 2023), while the frequency of internet use has been linked to volumetric changes of the putamen in young women (Altbacker et al., 2016). These findings together imply that engagement with screen-based activities may influence the structural characteristics of the putamen. Moreover, the putamen is associated with cognitive functions, including cognitive control, working memory, episodic memory, and category learning (Ell et al., 2011); therefore, we examined the associations between screen time, cognitive function, and putamen volume, and explored whether putamen volume may be involved in the relationship between screen time and cognitive function.

2. Materials and methods

2.1. Participants

The ABCD study is a large research project that follows 11,878 children who were 9–10 years old at the start. These children were recruited from 21 places in the United States (Jernigan et al., 2018). Each research site obtained ethical approval, and data collection adhered to set protocols. Before participation, parents provided consent, and children agreed after the study was explained to them (Garavan et al., 2018). For this analysis, we used demographic, behavioral, and brain imaging data from the ABCD Study's 5.0 release, available through the National Institute of Mental Health Data Archive. A detailed summary of the sample and results is presented in Table 1. All statistical models used the full dataset to enhance the reliability of the findings and its applicability to a wider group.

2.2. Screen time

Screen time was measured using a questionnaire in which participants reported the amount of time they spent on digital devices, including television, videos, video games, texting, and video chat. The ABCD study showed screen use for weekdays and weekends separately.

Table 1
The measures of five domains of cognitive function.

Domains of cognitive function	Measures in the National Institutes of Health (NIH) toolbox
Executive function	The average of scores in the dimensional change card sort test and the flanker inhibitory control and attention test
Processing speed	The score in the pattern comparison processing speed test
Working memory	The score in the list sorting working memory test
Episodic memory	The score in the picture sequence memory test
Language	The average of scores in the oral reading recognition and picture vocabulary tests

We calculated a daily average by using the following formula: $(5/7 \times \text{weekday screen time}) + (2/7 \times \text{weekend screen time})$ (Yang et al., 2023). And the screen time for each digital device was also calculated by this formula.

2.3. Cognitive function

Participants' cognitive function was measured with the National Institutes of Health (NIH) toolbox—an iPad-based program designed to assess cognitive function in individuals aged three years or older. The cognition battery comprised seven measures to assess five domains: executive function, processing speed, working memory, episodic memory, and language (Fox et al., 2021; Nolin et al., 2023; Ott et al., 2022; Yamashita et al., 2024). The measures for each domain are summarized in Table 1. We used the age-corrected scores of these indices for the analysis.

2.4. Putamen

Information about magnetic resonance imaging acquisition and data preprocessing in the ABCD study has been published (Casey et al., 2018; Hagler Jr. et al., 2019). Three Tesla magnetic resonance scanners (Siemens, Philips, and General Electric 750) were used in ABCD study to obtain high-resolution T1-weighted three-dimensional structural images with 1 mm isotropic. The acquisition parameters were as previously published. FreeSurfer (version 5.3.0) processed the structural data in the ABCD study using a standardized processing pipeline. In our previous study, the putamen was the only brain region associated with screen time (Shou et al., 2025); thus, this study extracted the putamen as a region of interest. In the ABCD study, the putamen is assessed with an atlas-based segmentation. Putamen volume is calculated by averaging the volumes of the right and left putamen. We analyzed the data from participants whose structural imaging data passed the FreeSurfer quality control check.

2.5. Demographic variables and covariates

We included the variables in Table 1 as covariates based on previous ABCD research (Bernanke et al., 2022; Hamatani et al., 2022; Hiraoka et al., 2023; Owens et al., 2021; Yamashita et al., 2024). Sex and race/ethnicity (White, Black, Hispanic, Asian, or other) were turned into dummy variables. Parental income was divided into five groups. Age, parental education level, pubertal status, total brain size, daily sleep time, and physical activity level were measured as continuous numbers. Parental education was noted by years of schooling. Pubertal status was checked using the Pubertal Development Scale (Petersen et al., 1988). Additionally, sleep duration and physical activity frequency, commonly considered when analyzing the association between screen-related behaviors and cognitive function, were included as covariates (Healy et al., 2020; Sisson et al., 2010). Demographic variables and covariates are summarized in Table 2.

2.6. Statistical analysis

We used R software (version 4.3.1) for our statistical analyses. We used the “lmerTest” package in R for the mixed-effect regression analysis and “lavan” R package for the standard three-variable mediation analysis. Outliers in screen time, putamen volume, cognitive function, and all continuous variables were adjusted to be within three standard deviations from the average. First, we examined the association between screen time and cognitive function. To study the link between screen time and cognitive function, we applied the linear mixed-effects model with each domain of cognitive function as the dependent variable and screen time as the independent variable. The covariates included children's age, sex, pubertal status, race, parental education, household income, physical activity and sleep duration. In line with previous studies

Table 2

Participants' demographic data and key study variables.

	Descriptive statistics	Number of participants (n)
Age (months) (mean $\pm$ SD)	119.00 $\pm$ 7.50	11,867
Sex		11,868
Male (n)	6188	
Female (n)	5677	
Parents' annual income (USD)		10,851
< 49,999 (n)	3222	
50,000–74,999 (n)	1498	
75,000–99,999 (n)	1570	
100,000–199,999 (n)	3311	
$\geq$ 200,000 (n)	1250	
Parents' education (years) (mean $\pm$ SD)	15.20 $\pm$ 2.61	11,153
Race		11,866
White (n)	6173	
Black (n)	1784	
Hispanic (n)	2410	
Asian (n)	252	
Other (n)	1247	
Pubertal status (score) (mean $\pm$ SD)	1.64 $\pm$ 0.42	11,783
Sleep duration (hours) (mean $\pm$ SD)	9.76 $\pm$ 1.27	11,863
Physical activity (times per week) (mean $\pm$ SD)	3.50 $\pm$ 2.32	11,840
Handedness-score rating		11,868
Right-handed (n)	9419	
Left-handed (n)	730	
Ambidextrous (n)	1361	
Total intracranial volume (cm ³) (mean $\pm$ SD)	1489.52 $\pm$ 144.00	11,728
Screen time (hour) (mean $\pm$ SD)	3.79 $\pm$ 3.07	11,843
Executive function (score) (mean $\pm$ SD)	96.07 $\pm$ 12.22	11,710
Processing speed (score) (mean $\pm$ SD)	93.78 $\pm$ 22.07	11,697
Working memory (score) (mean $\pm$ SD)	100.55 $\pm$ 14.79	11,672
Episodic memory (score) (mean $\pm$ SD)	100.96 $\pm$ 16.10	11,709
Language (score) (mean $\pm$ SD)	104.65 $\pm$ 15.62	11,702

(Hamatani et al., 2022; Hiraoka et al., 2023), the family ID (denoting sibling status), multiple data collections, and twin or triplet status were controlled as random effects. For the analysis of the relationship between screen time and cognitive function, we included participants with available demographic, screen time, and cognitive function data.

Second, we assessed whether putamen volume mediated the association between screen time and cognitive function. A mediation analysis was conducted for screen time, putamen volume, and the domains of cognitive function, which were significantly associated with screen time. Putamen volume and cognitive function were residualized for covariates with the linear mixed-effects model noted above and converted into z-scores. We estimated the significance of the mediating effect with a bias-corrected bootstrap approach with 10,000 random samples. For mediation analysis, we included participants with available demographic data, screen time, cognitive function data, and MRI structural data who passed the FreeSurfer quality control check in the ABCD study.

The significance level was set at 0.05, and since the analysis included different cognitive domains, the result was corrected for multiple comparisons using false discovery rate (FDR) adjustments within each domain (Hochberg & Benjamini, 1990).

3. Results

3.1. Association between screen time and cognitive function

We adopted linear mixed-effects regression models with each domain of cognitive function as the dependent variable and screen time

as the independent variable. Executive function ($b = -0.212$, 95% CI = -0.301 to -0.122 , $p < .001$, $FDR p < .001$, $\beta = -0.051$, $n = 9979$), processing speed ($b = -0.327$, 95% CI = -0.492 to -0.161 , $p < .001$, $FDR p < .001$, $\beta = -0.042$, $n = 9971$), language function ($b = -0.399$, 95% CI = -0.501 to -0.297 , $p < .001$, $FDR p < .001$, $\beta = -0.075$, $n = 9973$), working memory ($b = -0.396$, 95% CI = -0.501 to -0.290 , $p < .001$, $FDR p < .001$, $\beta = -0.078$, $n = 9954$) and episodic memory ($b = -0.361$, 95% CI = -0.479 to -0.244 , $p < .001$, $FDR p < .001$, $\beta = -0.066$, $n = 9979$) all showed significant associations with screen time. The results are summarized in Fig. 1. We also examined the relationship between screen time on each digital device and each domain of cognitive function, and summarized the results in supporting information (Table S1 and Fig. S1). We found that almost all of the screen time on each digital device is negatively associated with cognitive function, even after FDR correction. Therefore, in the following analyses, only the total time was used.

Additionally, we also examined the correlation among each domain of cognitive function and summarized the result in supporting information (Table S2). The result showed that there are high correlations among each domain of cognitive function.

3.2. Mediating effect of putamen volume on the relationship between screen time and cognitive function

We examined the mediating effects of putamen volume on the relationship between screen time and cognitive function. The results showed a significant partial mediation effect of putamen volume on the association between screen time and executive function (indirect effect: $\beta = -0.015$, 95% CI = -0.028 to -0.005 , $p = .016$, $FDR p = .027$; total effect: $\beta = -0.509$, 95% CI = -0.728 to -0.293 , $p < .001$, $FDR p < .001$, $n = 9369$), processing speed (indirect effect: $\beta = -0.012$, 95% CI = -0.024 to -0.004 , $p = .015$, $FDR p = .027$; total effect: $\beta = -0.306$, 95% CI = -0.506 to -0.110 , $p < .001$, $FDR p < .001$, $n = 9369$), and language (indirect effect: $\beta = -0.005$, 95% CI = -0.010 to -0.001 , $p = .016$, $FDR p = .027$; total effect: $\beta = -0.334$, 95% CI = -0.416 to -0.251 , $p < .001$, $FDR p < .001$, $n = 9343$). The results showed no significant mediation effect on the relationship between screen time and episodic memory (indirect effect: $\beta = -0.0006$, 95% CI = -0.006 to 0.006 , $p = .867$, $FDR p = .867$, total effect: $\beta = -0.403$, 95% CI = -0.535 to -0.269 , $p < .001$, $FDR p < .001$, $n = 9369$) and working memory (indirect effect: $\beta = -0.001$, 95% CI = -0.005 to 0.004 , $p = .759$, $FDR p = .867$; total effect: $\beta = -0.409$, 95% CI = -0.523 to -0.294 , $p < .001$, $FDR p < .001$, $n = 9369$). The results are summarized in Fig. 2.

4. Discussion

The findings highlight the association between screen time and cognitive function, with putamen volume potentially accounting for part of this relationship. This study analyzed data from approximately 10,000 children to examine this association using five domains measured with the NIH toolbox. Moreover, the putamen, a key component of the basal ganglia (Ghandili & Munakomi, 2024), and our results suggest that it may be involved in the association between screen time and cognitive function.

Our results revealed a significant negative association between screen time and overall cognitive function, including executive function, working memory, language, episodic memory, and processing speed, which aligns with prior results (Paulich et al., 2021; Shalash et al., 2024). Similarly, longer night screen time was linked to lower abilities in working memory, calculation, information-processing speed, and attention in adults (Shalash et al., 2024). These findings reinforce the idea that excessive screen time is adversely associated with various aspects of cognitive function across different age groups. Additionally, by using a mixed-effects model and controlling for various confounding factors, such as physical activity and sleep duration, the current results highlight the link between longer screen time and reduced cognitive

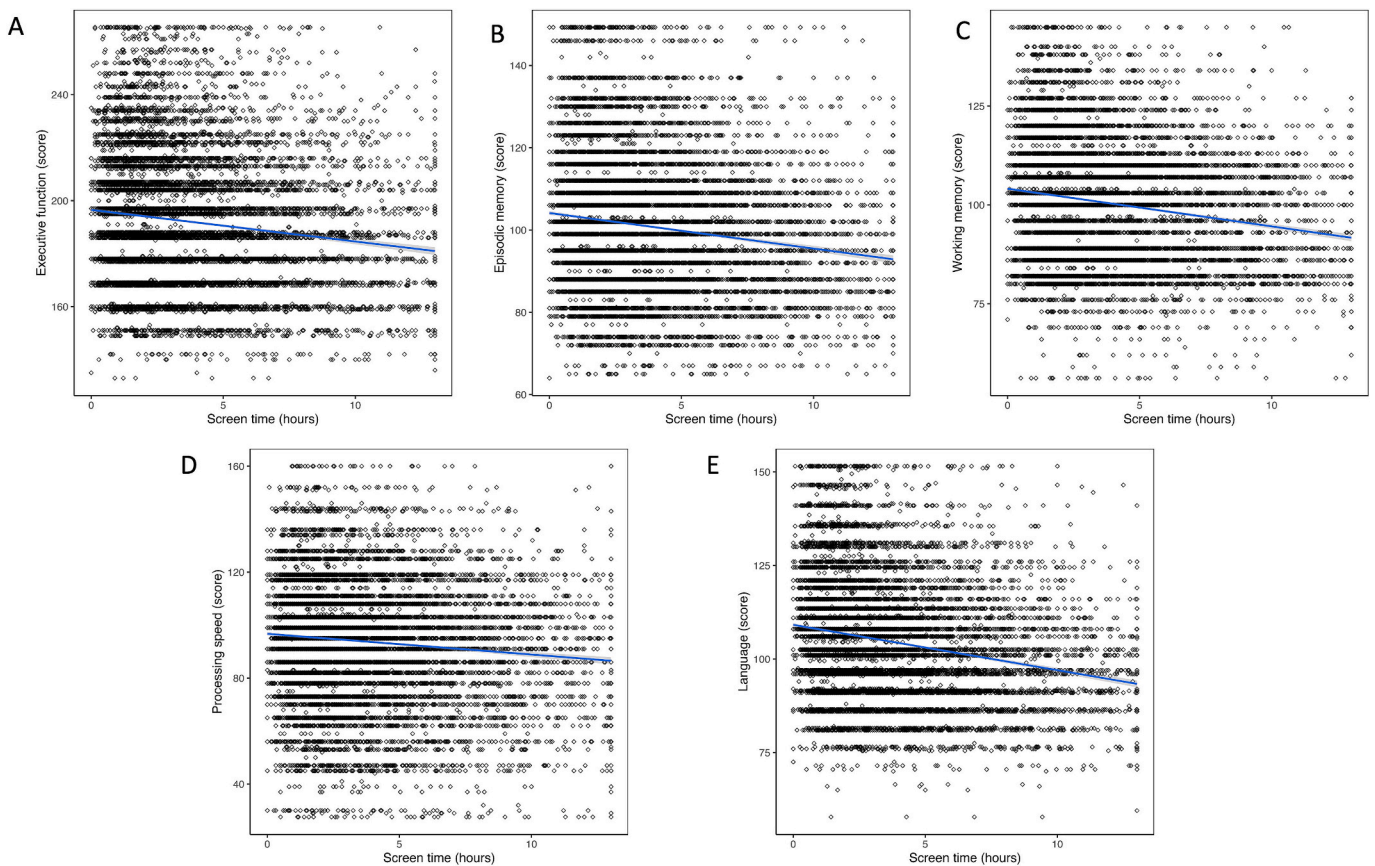


Fig. 1. The association between screen time and cognitive function. A) The association between screen time and executive function score; B) The association between screen time and episodic memory score; C) The association between screen time and working memory score; D) The association between screen time and processing speed score; E) The association between screen time and language score.

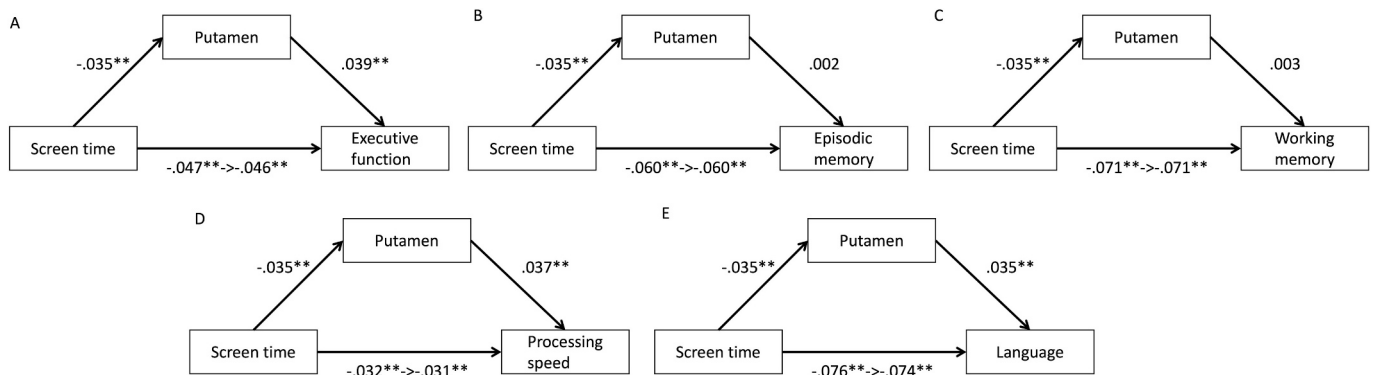


Fig. 2. The mediating effect of putamen volume on the association between screen time and cognitive function. A) The mediating effect of putamen volume on the association between screen time executive function score; B) The mediating effect of putamen volume on the association between screen time episodic memory score; C) The mediating effect of putamen volume on the association between screen time working memory score; D) The mediating effect of putamen volume on the association between screen time processing speed score; E) The mediating effect of putamen volume on the association between screen time language score.

function without sleep disturbance and physical activity. Previous studies (Cajochen et al., 2011; Faught et al., 2017) have indicated that sleep and physical activity are simultaneously associated with screen-related activity and cognitive function, and they also mediate the relationship between screen time and cognitive function. This finding emphasizes the significance of association of screen exposure with support healthy cognitive and neurological development during childhood. It is worth noting that, given the significant correlations among cognitive function domains, screen time may be associated with the shared component across these domains rather than with the unique variance of

each domain. Additionally, the cross-sectional design of the analysis limits our ability to infer a causal relationship between screen time and cognitive function.

The novel contribution of this study lies in identifying the partial mediating effect of putamen volume on the relationship between screen time and cognitive function, particularly executive function, processing speed, and language. The putamen is involved in both screen-related activity and cognitive function (Chen et al., 2023; Ell et al., 2011; Healy et al., 2020). Our findings indicate that the putamen may be partially involved in the association between screen time and executive

function, processing speed, and language. Based on the result, the correlation between screen time and cognitive function can be partly explained by putamen volume in the current study. One possible explanation is that screen-related activities may be associated with putamen function, and putamen-related differences may, in turn, be linked to cognitive function-related brain networks, potentially relating to cognitive function.

The possible neural mechanisms underlying the link between screen exposure and cognitive function may involve dopaminergic modulation and altered functional connectivity within fronto-striatal circuits. Previous studies showed that excessive screen exposure, particularly involving rewarding digital content, activates the mesolimbic dopamine system, including the nucleus accumbens and putamen (Chen et al., 2023; Kühn & Gallinat, 2015). Chen et al. (2023) found that children with more screen time showed a higher reward orientation and weaker fronto-striatal connectivity. Such repeated stimulation may lead to neural adaptations similar to those observed in behavioral addictions, including altered reward sensitivity and disrupted top-down control from the prefrontal cortex (Chen et al., 2023). Moreover, previous studies have shown that the structural and functional connectivity between the putamen and the prefrontal cortex directly influences cognitive functions, such as executive function (Ell et al., 2011) and language (Viñas-Guasch & Wu, 2017). This implies that dopaminergic activity in the putamen may potentially mediate the putamen's contribution to cognitive function (van Beilen & Leenders, 2006). Collectively, these findings suggest that the putamen serves as a key node linking screen exposures with cognitive outcomes. Structural and functional alterations in the putamen, particularly those affecting dopaminergic circuits and fronto-striatal connectivity, provide a plausible neurobiological mechanism by which screen time may impact cognitive function in children. Therefore, the partial mediating role of the putamen observed in this study might reflect broader alterations in dopaminergic signaling and cortico-striatal network organization associated with screen exposure. Future studies should consider examining the role of the cortico-striatal network in the relationship between screen time and cognitive function.

Compared with previous studies, our findings both support and extend prior evidence. For example, prior research has demonstrated that increased screen time is associated with reduced cognitive performance and altered striatal activity in children (Paulich et al., 2021; Walsh et al., 2020), but few studies have explicitly identified the structural brain as a partial mediator linking screen use and specific cognitive domains such as executive function, processing speed, and language. Our results provide preliminary insight into potential neural substrates of this association and are broadly consistent with prior evidence indicating that the putamen is involved in cognitive regulation and may be associated with environmental factors, such as screen exposure.

This study has several limitations. First, as a cross-sectional study, it cannot establish causal relationships. Although we highlighted the possibility that excessive screen time may affect brain structure and cognitive function, it is also plausible that pre-existing cognitive deficits lead to increased screen time, which in turn may influence putamen volume. Second, because this study was based on an existing database, we were unable to collect data independently. As a result, the study is subject to limitations such as self-report bias, scanner heterogeneity, and other unmeasured confounding factors. In addition, substantial shared variance among cognitive domains was observed, making it difficult to disentangle the relationships between screen time and specific cognitive functions. These limitations highlight the need for further research to explore the long-term implications of screen time on brain development and cognitive health. Longitudinal studies tracking changes over time are essential for understanding the potential reversibility of these effects and identifying critical periods for intervention.

5. Conclusion

This study demonstrates a negative association between excessive screen time and cognitive function in children, with the putamen potentially involved in this relationship. These findings contributed to a better understanding of the neural correlates associated with screen time and cognitive function. Although causal inferences cannot be drawn, the results may inform future longitudinal and interventional studies aimed at promoting healthier screen use and supporting cognitive and brain development.

CRedit authorship contribution statement

Qiulu Shou: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Masatoshi Yamashita:** Writing – review & editing, Validation, Software, Methodology, Funding acquisition, Data curation. **Yoshifumi Mizuno:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflicts of interest related to this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2026.106742>.

Data availability

The data for this study came from the ABCD study (<https://abcdstudy.org/>) in the NIMH Data Archive. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A list of participating sites and a complete list of study investigators can be found at https://abcdstudy.org/consortium_members/.

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Glossary

Cognitive function: Performance of the mental processes of working memory, language, executive function, processing speed, and episodic memory

Screen time: The total duration spent on various devices, including playing video games and watching television in a single day